

SQL Queries, Business Questions, Insights & Concepts

A. KPI's

1. Total Revenue

Business Question

What is the total revenue generated from pizza sales?

Query

```
SELECT SUM(total_price) AS total_revenue  
FROM pizza_sales;
```

Concepts Used

- `SUM()`
- Aggregate Function

Insight Derived

Helps measure the overall business revenue generated from pizza sales.

2. Average Order Value

Business Question

What is the average amount spent per order?

Query

```
SELECT ROUND(SUM(total_price) / COUNT(DISTINCT order_id), 2) AS avg_order_value  
FROM pizza_sales;
```

Concepts Used

- `SUM()`
- `COUNT()`

- `DISTINCT`
- `ROUND()`

Insight Derived

Shows customer spending behavior and average transaction value.

3. Total Pizzas Sold

Business Question

How many pizzas were sold overall?

Query

```
SELECT SUM(quantity) AS total_pizzas_sold  
FROM pizza_sales;
```

Concepts Used

- `SUM()`

Insight Derived

Measures total sales volume and customer demand.

4. Total Orders

Business Question

How many orders were placed by customers?

Query

```
SELECT COUNT(DISTINCT order_id) AS total_orders  
FROM pizza_sales;
```

Concepts Used

- `COUNT()`
- `DISTINCT`

Insight Derived

Represents total customer transactions during the sales period.

5. Average Pizzas Per Order

Business Question

On average, how many pizzas are ordered per transaction?

Query

```
SELECT ROUND(SUM(quantity) / COUNT(DISTINCT order_id), 2) AS avg_pizzas_per_order
FROM pizza_sales;
```

Concepts Used

- SUM()
- COUNT()
- ROUND()

Insight Derived

Helps understand customer ordering patterns and basket size.

B. DAILY TREND FOR TOTAL ORDERS

Business Question

Which days receive the highest number of customer orders?

Query

```
SELECT DAYNAME(order_date) AS order_day,
COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
```

GROUP BY DAYNAME(order_date)

ORDER BY total_orders DESC;

Concepts Used

- DAYNAME()
- COUNT()
- GROUP BY
- ORDER BY

Insight Derived

Identifies peak business days and customer ordering behavior throughout the week.

C. HOURLY TREND FOR ORDERS

Business Question

At which hours are customer orders highest?

Query

```
SELECT HOUR(order_time) AS order_hour,  
COUNT(DISTINCT order_id) AS total_orders  
FROM pizza_sales  
GROUP BY HOUR(order_time)  
ORDER BY order_hour;
```

Concepts Used

- HOUR()
- COUNT()
- GROUP BY
- ORDER BY

Insight Derived

Helps identify peak ordering hours for staffing and operational planning.

D. % OF SALES BY PIZZA CATEGORY

Business Question

Which pizza categories contribute the most to total revenue?

Query

```
SELECT pizza_category,  
ROUND(SUM(total_price), 2) AS total_revenue,  
ROUND(SUM(total_price) * 100 /  
(SELECT SUM(total_price) FROM pizza_sales), 2) AS pct_sales  
FROM pizza_sales  
GROUP BY pizza_category  
ORDER BY pct_sales DESC;
```

Concepts Used

- GROUP BY
- Subquery
- SUM()
- ROUND()

Insight Derived

Shows the revenue contribution of each pizza category and identifies the most profitable categories.

E. % OF SALES BY PIZZA SIZE

Business Question

Which pizza sizes generate the highest sales revenue?

Query

```
SELECT pizza_size,  
  
ROUND(SUM(total_price), 2) AS total_revenue,  
  
ROUND(SUM(total_price) * 100 /  
  
(SELECT SUM(total_price) FROM pizza_sales), 2) AS pct_sales  
  
FROM pizza_sales  
  
GROUP BY pizza_size  
  
ORDER BY pct_sales DESC;
```

Concepts Used

- GROUP BY
- Subquery
- SUM()
- ROUND()

Insight Derived

Helps understand customer preference for pizza sizes and their revenue impact.

F. TOTAL PIZZAS SOLD BY PIZZA CATEGORY

Business Question

Which pizza categories sold the highest quantity in February?

Query

```
SELECT pizza_category,  
  
SUM(quantity) AS total_quantity_sold  
  
FROM pizza_sales  
  
WHERE MONTH(order_date) = 2  
  
GROUP BY pizza_category  
  
ORDER BY total_quantity_sold DESC;
```

Concepts Used

- WHERE
- MONTH()
- GROUP BY
- SUM()

Insight Derived

Helps compare category-wise sales performance during a selected month.

G. TOP 5 BEST SELLERS BY TOTAL PIZZAS SOLD

Business Question

Which pizzas are the top-selling products?

Query

```
SELECT pizza_name,  
  
SUM(quantity) AS total_pizza_sold  
  
FROM pizza_sales
```

GROUP BY pizza_name

ORDER BY total_pizza_sold DESC

LIMIT 5;

Concepts Used

- GROUP BY
- ORDER BY
- LIMIT
- SUM()

Insight Derived

Identifies the most popular pizzas contributing the highest sales volume.

H. BOTTOM 5 WORST SELLERS BY TOTAL PIZZAS SOLD

Business Question

Which pizzas have the lowest sales performance?

Query

SELECT pizza_name,

SUM(quantity) AS total_pizza_sold

FROM pizza_sales

GROUP BY pizza_name

ORDER BY total_pizza_sold ASC

LIMIT 5;

Concepts Used

- GROUP BY
- ORDER BY
- LIMIT
- SUM()

Insight Derived

Highlights underperforming pizzas that may require marketing or menu optimization.

FILTER-BASED ANALYSIS

Monthly Filter Example

Business Question

How do order trends behave during January?

Query

```
SELECT DAYNAME(order_date) AS order_day,  
COUNT(DISTINCT order_id) AS total_orders  
FROM pizza_sales  
WHERE MONTH(order_date) = 1  
GROUP BY DAYNAME(order_date);
```

Concepts Used

- WHERE
- MONTH()
- GROUP BY

Insight Derived

Supports monthly sales analysis and helps identify seasonal demand patterns.

Quarterly Filter Example

Business Question

How does customer ordering behavior vary during Quarter 1?

Query

```
SELECT DAYNAME(order_date) AS order_day,  
COUNT(DISTINCT order_id) AS total_orders  
FROM pizza_sales  
WHERE QUARTER(order_date) = 1  
GROUP BY DAYNAME(order_date);
```

Concepts Used

- `QUARTER()`
- `GROUP BY`
- `COUNT()`

Insight Derived

Helps analyze quarterly customer demand and business performance trends.